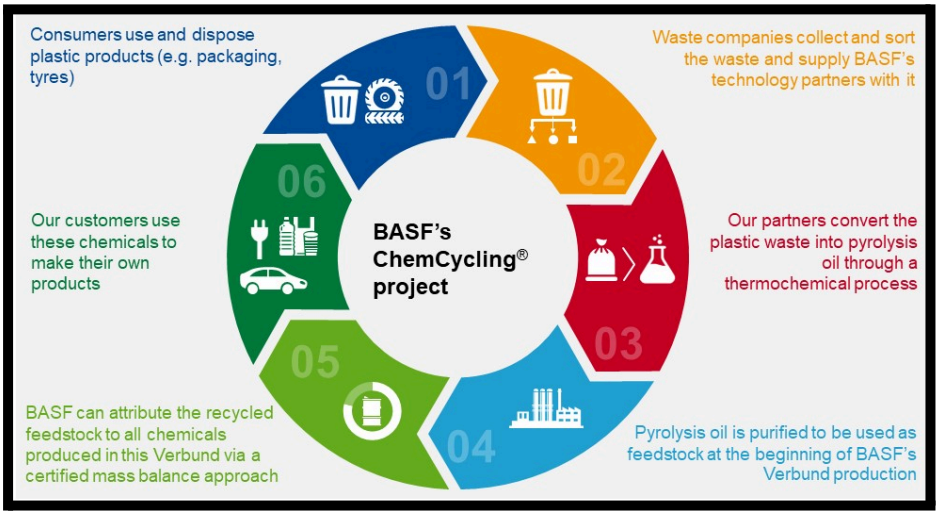
Company Overview: BASF SE

BASF SE is a German chemical company and one of the world's largest chemical producers. The company has been around since 1865 and is based in Ludwigshafen, Germany. Today, BASF has operations in more than 80 countries and supplies products and solutions for key markets such as automotive, construction, agriculture, and consumer business.

ESG Focus: BASF

Environmental: The ChemCycling Project



Why (Problem)

The problem with conventional mechanical recycling is that it cannot recycle "dirty" plastics, mixed plastics, or tires. In addition, mechanical recycling eventually deteriorates the plastic. Most of this waste is currently being burned as energy or put into landfills, producing CO2 emissions and squandering valuable carbon molecules.

How (ChemCycling)



BASF has developed ChemCycling, a proprietary thermochemical process known as ***pyrolysis***. Instead of merely washing and melting the plastic, they are breaking it down at the molecular level by applying high temperatures in the absence of oxygen. The plastic is now changed into ***pyrolysis oil.*** BASF is now using this oil at the very beginning of their massive production facility, known as the Verbund site. This oil is now used instead of the fossil-based naphtha, i.e., crude oil, used in the production of new chemicals.

What (Outcome)

This leads to the creation of "Ccycled®" products. This plastic, having been recycled to a molecular level, makes the products equivalent in quality to those made from virgin oil.

Impact: BASF plans to process 250,000 metric tons of recycled and waste-based raw materials annually by the year 2026. This process will reduce the dependence of the company on foreign fossil fuels and decrease the carbon footprint of the process.